

EFlux: A Hierarchical LLM - RL Collaboration Framework for Algorithmic Electricity Trading by Virtual Power Plants

EFlux: 面向虚拟电厂算法电力交易的层次化 LLM - RL 协同框架

01 Multi-Timescale Hierarchical LLM - RL Collaboration

多时间尺度的层次化 LLM - RL 协同决策

中文 提出由低频 LLM 战略层与高频 RL 执行层组成的协同架构。LLM 负责市场状态理解、风险姿态和长期目标调节，RL 负责结构化动作空间内的实时交易决策，并将执行结果反馈给战略层。

English We propose a multi-timescale hierarchical LLM - RL architecture in which a slow LLM strategist performs market reasoning and strategic control, while a fast RL policy makes real-time tactical decisions and returns execution feedback to the strategic layer.

02 Human-Guided Semantic Shared Control

人在环路的语义化共享控制机制

中文 交易员可通过风险预算、目标 SOC、策略偏好、暂停和临时接管等高层语义对智能体进行监督与干预，而无需逐笔构造底层订单；干预结束后，控制权可交还给自主策略。

English We introduce a human-guided semantic shared-control mechanism that enables operators to supervise, steer, and temporarily override the autonomous agent through bounded high-level trading intentions rather than raw market orders.

03 Physics-Constrained Intent Compilation and Auditable Execution

多源意图的物理约束编译与可审计执行

中文 将人类指导、LLM 战略意图和 RL 战术动作统一映射为可解释的交易原语，再经确定性编译和风险网关生成满足 SOC、功率、能源、现金及交付约束的订单，并完整记录从指导到结算的决策来源。

English We develop a physics-constrained intent compilation and auditing mechanism that transforms human, LLM, and RL decisions into safe executable orders while preserving end-to-end provenance from guidance and risk validation to delivery and settlement.

04 Open, Extensible, and Reproducible VPP Trading Testbed

开放、可扩展、可复现的虚拟电厂交易测试平台

中文 提供面向研究人员和开发者的统一测试平台。内置与外部智能体可通过 API、SDK 或智能体协议接入同一市场、物理、风控、交付和结算链路，并利用标准场景、基准策略和确定性回放进行公平比较。

English We provide an open, extensible, and reproducible testbed in which built-in and external agents access the same market, physical, risk, delivery, and settlement pipeline through a unified protocol, enabling standardized testing and fair comparison.